

(185)

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Exam 4: NE533: Nuclear Fuel Performance

Show all work. Label question number in your response. Pay attention to units. Point values correspond to expected depth of response.

1. (18 pts) A ZIRLO cladding tube is in reactor at 625 K for 500 days. The initial wall thickness is 600 μm .
 - a) Estimate the oxide thickness after this time?
 - b) Assuming the hydrogen pickup fraction is 14%, what is the weight ppm of hydrogen in the cladding? Assume $\text{PBR} = 1.56$, $\rho_{\text{Zr}} = 6.5 \text{ g/cc}$, $\rho_{\text{ZrO}_2} = 5.68 \text{ g/cc}$.
2. (2 pts) What is the rate-limiting step in the aqueous corrosion of Zr cladding?
3. (8 pts) Where do hydrides form in the cladding? Why? What impacts do hydrides have?
4. (12 pts) What is a RIA? What is a typical RIA in a PWR or BWR. Describe what happens during a RIA.
5. (12 pts) What is a LOCA? Describe the impacts of a LOCA. How is it different than a RIA?
6. (4 pts) How does burnup impact the type of failure and the failure probability during an accident?
7. (6 pts) What are the four pathways to make the fuel/cladding system more accident tolerant?
8. (6 pts) List one near-term ATF concept and why it is of interest.
9. (6 pts) What happens to zirconium cladding when it is exposed to high temperature steam?
10. (6 pts) List three examples of limiting phenomena governing LWR operation.
11. (6 pts) What fuel performance and safety impacts does CRUD have?
12. (6 pts) List two water chemistry controls that have been implemented in LWRs, including why they were implemented.
13. (8 pts) List and describe two key microstructural phenomena that are unique to MOX fuels.

1.) $T = 625\text{K}$
 $t = 500\text{ days}$

$\delta^0 = 600\mu\text{m}$

a.) $t^*(d) = 6.62 \times 10^{-7} \exp\left(\frac{11949}{T}\right)$
 $= 6.62 \times 10^{-7} \exp\left(\frac{11949}{625}\right)$

16/18

$t^*(d) = 133\text{ days}$ -

$t^*(d) < t$

$\delta^*(\mu\text{m}) = 5.1 \exp\left(\frac{-550}{T}\right)$ -
 $= 5.1 \exp\left(\frac{-550}{625}\right)$

$\delta^*(\mu\text{m}) = 2.115\mu\text{m}$ -

$K_L\left(\frac{\mu\text{m}}{d}\right) = 7.48 \times 10^6 \exp\left(\frac{-12500}{T}\right)$ -
 $= 7.48 \times 10^6 \exp\left(\frac{-12500}{625}\right)$

$K_L\left(\frac{\mu\text{m}}{d}\right) = 0.0154$ -

$\delta(\mu\text{m}) = \delta^* + K_L(t - t^*)$ -

$= 2.115\mu\text{m} + (0.0154)(500 - 133)$

$\delta(\mu\text{m}) = 7.77\mu\text{m}$ -
 ANS

b.) $f = 14\%$

$$C_H^{clad} (\text{wt. ppm}) = \frac{2f \times \delta \times \rho_{\text{oxid}} \times f_{\text{ZrO}_2}^0 \times \frac{M_H}{M_O}}{\left(t - \frac{\delta}{\rho_{\text{PBE}}}\right) \times \rho_{\text{metal}}} \times 10^6$$

$$f_{\text{ZrO}_2}^0 = \frac{16}{(2 \cdot 16) + 91} = 0.13 \quad \frac{32}{32 + 91} = 0.26$$

$$\frac{M_H}{M_O} = \frac{1}{16} \quad \checkmark$$

$$C_H^{clad} (\text{wt. ppm}) = \frac{2(0.14)(7.77 \text{ mm})(5.68 \frac{\text{g}}{\text{cm}^3})(0.13)(\frac{1}{16})}{\left(\underset{400}{500} - \frac{7.77 \text{ mm}}{1.56}\right)(6.5 \frac{\text{g}}{\text{cm}^3})} \times 10^6$$

$$C_H^{clad} (\text{wt. ppm}) = 4.02 \text{ wt. ppm}$$

ANS

2.) The formation of the oxide layer. The thickness of the oxide layer limits the amount of "new" Zirconium that can be reduced, simply because a growing oxide layer exists. As oxidation occurs the layer limits further layers from forming.

-kind of...

3.) Hydrides form in the bulk but move to places like crack tips.

They move this way due to diffusion caused by the temperature gradient in the fuel/clad system. Hydrides can cause IGSCC and cause cracks to further propagate and cause mechanical failure.

4.) A RIA is a Reactivity Insertion Accident. A typical RIA for a PWR is a control rod ejection and for a BWR a control rod drop. During a RIA there is an increase in reactivity which causes a power increase, and power hot spots, which cause stress on core components depending on the pulse of the RIA. RIAs also cause

-needed more

5.) A LOCA is a Loss of Cooling Accident. A LOCA is different than a RIA due to the effects causing it. A LOCA occurs when there is a cooling channel blockage, obstruction, or even Departure from Nucleate Boiling. In both cases, the coolant is not effectively removing heat from the fuel assemblies and can cause fuel failure, assembly bow, boiling crisis, and others.

-missed some

6.) Burnup impacts the type of failure due to the structure of the fuel pellet. As burnup increases, the pellet swells due to radiation induced voids, fission gas build-up, etc. As burnup increases, the gap between the fuel pellet and the clad closes. Once closed, mechanical stress →

-cladding embrittled

6. Continued) is present in the clad. Over time as burnup increases, the clad may rupture leading to potential LOCA accidents.

7.) Oxidation - Resistant clad/components.

(Increase Heat Transport of fuel/clad. - kind of?)

4/6 ^{Same thing} Corrosion - Resistant clad/components.

Increased Fuel Characteristics (Swelling reduction, FP retention, TC increase, etc.)

8.) The use of coated cladding. Coated cladding has manufactured materials on the cladding that increase corrosion resistance that affect current cladding. These advanced coated claddings can be used very soon as they do not greatly reduce neutron economy, thermal transport, mechanical viability, etc.

6/6 9.) Zr cladding may change phase depending on the temperature of the cladding. High temperature steam increases the oxidation of Zr cladding, therefore increasing the amount of Hydrogen that is being produced. ^{-what contrib.?}



This 2H_2 contributes in a number of ways, most notably as a flammable hazard in-core.

5/6 10.) The limit on fuel temperature due to Zr-cladding. ^{mostly} Above 800°C Zr can become a mix of α and β -Zr, and operation above $1,204^\circ\text{C}$ is not allowed. Due to Zr-cladding design, there is no room for higher temperature fuel efficiency like in advertised advanced reactor systems with LHRs above 400 w/cm . Another limit is fuel burnup levels. In typical LWRs burnup causes fuel pellets to significantly swell and keel. $\rightarrow$

10. continued)

Service life low (compared to advanced reactor system concepts).
With typical single-composition fuel there are limitations to how high of a level of burnup you can achieve, compared to MOX fuels (80% U, 20% Pu) that can achieve higher burnups (~15%). ~~not really~~

Another limitation is the oxidation-resistance of LWR [✓]cladding. Due to LWRs using water as a coolant, oxidation is an ever-present factor that limits the total output of an LWR. Compared to advanced systems like Sodium-Cooled Fast Reactors where oxidation on cladding is significantly less of a concern.

11.) CRUD impacts fuel performance and safety significantly. As CRUD is deposited onto (fuel) assemblies, where activation occurs and the activated products within the CRUD may dislodge and carry activated species further along into the primary system, notably the heat exchangers. Once in the heat exchangers and really anywhere in the primary system, plant personnel in the vicinity of these activated products can be affected by the dose they receive. CRUD also reduces the thermal transport of fuel assemblies by insulating / blocking coolant from contacting the cladding. CRUD may also flake off and cause channel blockages, obstruction, and further affect normal system operation.

12.) The changing / altering of primary system pH to more alkaline levels by the use of LiOH and Boric Acid. The more basic pH reduces the corrosion levels in the primary system that effectively lengthens the service life of components such as cladding and structural components. The addition of Aqueous Zinc into the primary system also reduces the activated products in the primary system. Since zinc is a "noble metal" it can be used to reduce ^{60}Co production from the stainless steel in the primary system, which further reduces doses to plant personnel and other system components.

13.) The first microstructural phenomena in MOX fuel is the change in grain structure axially within the pellet. As burnup increases, a central void forms due to pores and voids consolidating due to the fuel centerline temperature. With this void present, the grain structure changes axially. Columnar grains are oriented towards ^{-radially} the central void, where further towards the edge of the pellet there are equiaxed grains that resemble nominal grain growth seen in typical LWR-fuels. Then near the edge of the pellet is the as-sintered grain structure.

The second microstructural phenomena in MOX fuels is the formation of Joint Oxide Gains (JOG) fueled by the grain structure. As the pellet swells and eventually interacts with the clad, a small gap exists between the uneven clad-pellet surface which causes JOG products like CsMoO_4 to diffuse to and gather in the small boundary.